



Anna Kuriakose

# OWN CLOUD USING RASPBERRY

Public and Private Cloud Storage: A Practical Comparison

Cloud Based Software Engineering

2025

VAASAN AMMATTIKORKEAKOULU  
UNIVERSITY OF APPLIED SCIENCES (font size 11)  
Name of the Degree Programme (e.g. Energy Technology/ Information  
Technology/ International Business, font Verdana size 11)

## **ABSTRACT**

---

|                    |                                               |
|--------------------|-----------------------------------------------|
| Author             | Anna Kuriakose                                |
| Title              | OWN CLOUD USING RASPBERRY : Possible subtitle |
| Year               | 202X                                          |
| Language           | Finnish                                       |
| Pages              | 43 + 3 Appendices                             |
| Name of Supervisor | First name last name                          |

The abstract is written here using single spacing. The abstract is written in full-syllabic words. No hyphenation, underlining, italics or spacing is used. The abstract should be between 75 and 200 words in length. It is advisable to divide the text into paragraphs for readability, for example according to the following formula:

The first paragraph briefly describes the background to the research, the research problem and the task of the research.

The second paragraph briefly describes the theoretical framework of the study, the key concepts, the methodology and the research design used in the study.

The third chapter briefly presents the main findings and results of the study, as well as the main conclusions and interpretations.

Note: Only Finnish students include an abstract in Finnish if the thesis is written in English.

Use Finto (n.d.) to find keywords.

---

Keywords    learning, stimulation, teaching methods

# CONTENTS

|                                             |    |
|---------------------------------------------|----|
| ABSTRACT .....                              | 2  |
| 1 INTRODUCTION .....                        | 6  |
| 1.1 Background .....                        | 7  |
| 1.2 Motivations .....                       | 9  |
| 1.3 Objectives .....                        | 10 |
| 1.4 Scope of the Thesis .....               | 11 |
| 2 RELEVANT TECHNOLOGIES .....               | 12 |
| 2.1 Cloud Computing .....                   | 12 |
| 2.2 Consumer Choices of Cloud .....         | 13 |
| 2.3 Comparing CSPs .....                    | 14 |
| 2.4 Challenges and Benefits of CSPs .....   | 15 |
| 2.5 Raspberry Pi .....                      | 16 |
| 2.6 ownCloud .....                          | 17 |
| 2.7 User Concerns .....                     | 18 |
| 2.8 Hardware requirements .....             | 18 |
| 2.9 Software requirements .....             | 19 |
| 3 APPLICATION DESCRIPTION .....             | 21 |
| 3.1 QFD (Quality Function Deployment) ..... | 22 |
| 3.2 Functional Definition .....             | 23 |
| 3.3 Use Case Diagram .....                  | 25 |
| 3.4 Class Diagram .....                     | 26 |
| 3.5 Sequence Diagram .....                  | 29 |
| 3.6 Component Diagram .....                 | 30 |
| 3.7 Architectural Diagram .....             | 32 |
| 4 IMPLEMENTATION .....                      | 34 |
| 4.1 Implementation of cloud server .....    | 35 |
| 4.2 Installation and set up .....           | 36 |
| 5 TESTING .....                             | 40 |
| 5.1 Test Cases .....                        | 41 |
| 5.2 Results of Test Cases .....             | 43 |
| 5.3 Possible Improvements .....             | 45 |

|                 |    |
|-----------------|----|
| REFERENCES..... | 47 |
|-----------------|----|

## FIGURES

|                                                                                                        |    |
|--------------------------------------------------------------------------------------------------------|----|
| Figure 1. Types of cloud attacks ( <a href="http://www.aquasec.com">www.aquasec.com</a> , 2024). ..... | 8  |
| Figure 2. Use-case diagram .....                                                                       | 24 |
| Figure 3. Class diagram.....                                                                           | 26 |
| Figure 4. Sequence diagram. ....                                                                       | 28 |
| Figure 5. Component diagram. ....                                                                      | 30 |
| Figure 6. Architectural diagram. ....                                                                  | 32 |
| Figure 7. Raspberry Pi 3b. ....                                                                        | 34 |

## TABLES

|                                        |    |
|----------------------------------------|----|
| Table 1. Comparison of major CSPs..... | 13 |
| Table 1. Requirements Analysis. ....   | 21 |

## ABBREVIATIONS

|       |                                            |
|-------|--------------------------------------------|
| AMCIS | Americas Conference on Information Systems |
| CSP   | Cloud Service Provider                     |
| SBC   | Single Board Computer                      |
| SME   | Small and Medium Enterprise                |
| SSD   | Solid-State Drive                          |
| QFD   | Quality Function Deployment                |

## 1 INTRODUCTION

The internet and technology have become common among people of all ages in today's world. With the increasing number of internet users and advancements in technology, internet services are also becoming popular. A cloud service is one of many services. While most internet users may feel like the cloud is something that has developed and acquired popularity recently, looking at the history of cloud computing, the term "cloud" is traced back more than 60 years.

Thus, to keep up with the expanding "cloud", basic knowledge of the concept of cloud computing is necessary. Nonetheless, for the people outside of IT, the idea of cloud computing remains complex and unclear. This research aims to get the opinions of individuals and start up owners and people who work on cloud services. I collected the data through surveys, the example here describes what type of questions were asked and where, when, and how the survey was conducted. I created an online form with questions like occupation of the participant, types of devices used by him, amount storage utilized for personal or professional needs, amount of money spending on cloud services and opinion about owning a private cloud.

Emerging cloud service providers shows the need of cloud computing in the modern world. Cloud computing refers to a technology which allows a persons data to be kept online and be retrievable by that person from any location. Almost all the applications of IoT makes use of cloud computing and enrichment finance cloud hundreds of users holds the data and at a price offers this service to the user. What if there was your own cloud and you could put whatever you wanted anywhere in that cloud storage drive back from the world? I am showcasing how to build our own cloud using a Raspberry pi and a hard drive. For creating the cloud platform, I used ownCloud. There are many organizations that provide client server suit for creating file hosting services.

## **1.1 Background**

Hentschel et al. (2018) define cloud computing as “shared pool of managed and scalable IT resources on a rental basis” and elaborate that the definitions of cloud computing usually include the five main characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service. Moreover, most definitions of cloud computing include the three service models: Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), and Infrastructure-as-a-Service (IaaS). These services are offered through the four cloud computing deployment models: private, public, community, and hybrid. The definitions include resources such as networks, servers, storage, and applications that are offered in a scalable way through the internet without longterm capital expenditures or specific IT knowledge on the customer side (Hentschel et al., 2018).

With the technological growth of cloud computing, web applications have migrated towards clouds and have raised the concerns for privacy and security of user specific sensitive data, like how can an end user or consumers verify that a service provider conform to the privacy laws and protect consumers digital identity (Nida et al., 2014). Many businesses, such as those in manufacturing, research and the public sector, largely rely on internet technologies for data sharing and collaboration; this reliance is important in our linked world. Cloud solutions that are highly efficient and demonstrably secure are therefore analytically necessary. Their diverse needs often result in considerably different feature, administrative, security and cost requirements. Public cloud providers offer cost-effectiveness, high availability and scalability; however, important long-term costs are possible.

As public cloud offers various appealing benefits, there are a few disadvantages that comes along. The major downsides of public cloud computing are higher data security and privacy risk, performance and flexibility. Users of public cloud services usually have limited visibility

and less or no control over data security and privacy since the services and infrastructure are provisioned from the cloud service provider's data center. Users are not offered a guaranteed way to control and authorize access to their data in public cloud. Shared resources between multiple public cloud tenants bring up more vulnerabilities making security risks more intense. Compared to private cloud, network performance is less reliable in public cloud. Moreover, public cloud platforms are less flexible since they limit the customization of services and resources.

Sheps (2024) stated that, it has been recorded that there were 10 primary issues with security and privacy in public cloud, which stated in the below figure 1.



Figure 1. Types of cloud attacks (www.aquasec.com, 2024).

The ten security and privacy risks in public cloud computing mentioned above shows that users have lack of confidence and visibility when it comes to data security in public cloud.



## 1.2 Motivations

"Cloud computing can and does mean different things to different people. The common characteristics most interpretations share are on-demand scalability of highly available and reliable pooled computing resources, secure access to metered services from nearly anywhere, and displacement of data and services from inside to outside the organization. While aspects of these characteristics have been realized to a certain extent, cloud computing remains a work in progress" (Jansen & Grance, 2011). The main motivation of this research is to provide a scalable, secure, efficient and budget friendly cloud storage solution for customers in a long run.

This research discusses the need of private cloud storage among small scale business owners, individuals and for prototype building. Is renting a third party cloud service provider an efficient way for the above use cases? The responses from this industry specific people are collected and analysed through the survey. They have been introduced to the personal cloud storage using Raspberry Pi and collected opinions about using it in the future. Spreading an awareness of using cloud storage in an efficient way is another aim of this study. People who rent CSPs should be aware about the money they are spending in a long term.

Cloud users can access to the applications, and the data contained in the cloud from any location and at all times; data security and privacy issues in cloud computing are often concerning. Especially in a public cloud situation, security risks and problems are becoming more common. The total amount of cloud computing utilization, especially of public cloud utilization, has been growing fast in recent years, so a lot more sensitive data may be at risk. In accordance with Heiser (2014), this truly is a transition period in cloud computing during which the specific focus is sharply shifting from the cloud service provider to the cloud user. The statement has thoroughly prompted the issue of whether cloud

customers or cloud service providers are more responsible for data security as well as how cloud users can improve their own data protection (Ngo, 2018).

### **1.3 Objectives**

Choosing a cloud solution depends on its intended purpose, how it will be used and the kind of data it will process. This research points out the value of this assessment. It shows the practical and helpful nature of a Raspberry Pi and OwnCloud-based private cloud solution. This study examines performance, security, as well as cost-effectiveness to aid people along with organizations contemplating a transition to more secure and cost-efficient cloud storage.

This research will also explore the practical aspects of implementing, as well as managing, a private cloud solution. This covers the setup process, configuration options and common user challenges. This thorough research is about constructing and maintaining a private cloud, thus empowering users to make deeply informed decisions regarding their cloud storage requirements and effectively use the advantages of a Raspberry Pi along with an OwnCloud-based solution.

The main objectives of the research are:

- To design and set up a functional non-public cloud prototype using Raspberry Pi, hard drives, and ownCloud.
- Evaluate the device's usability, cost effectiveness, and scalability.
- Analyze the perceptions and desires of targeted users through a qualitative survey.

The research seeks to give readers more comprehension on cloud computing as well as data security problems in cloud computing, in addition to giving them a private cloud answer to protect their data as they enjoy cloud computing's benefits. For furnishing a solution for truly securing

data throughout the cloud, the thesis will particularly focus on comparing data security solutions between a public cloud and a private cloud. The thesis's main purpose involves providing a low budget, integrated and secured cloud solution without any involvement from a third-party. The solution is perfectly applied, and it is ideal for individual use. The thesis will provide detailed information on cloud computing and the latest issues in data security and privacy in cloud computing.

#### **1.4 Scope of the Thesis**

This research explores creating and evaluating a private cloud storage system with a Raspberry Pi and OwnCloud. This offers a cheaper data storage method for many small businesses and people. It also provides a considerably safer alternative to typical cloud services. This project uses the inexpensive Raspberry Pi and the open-source ownCloud to help users manage their data more effectively and cheaply.

The tasks of the thesis consist of two major parts, notably researching and implementing a private cloud. The research phase entirely involves seeking general details on cloud computing, individuals opinion on private and public cloud storage, present problems regarding data security and privacy in cloud computing, and the existing methods for protecting data in public and private clouds. The research additionally gives the reader information with respect to the cost that is involved in using cloud storage, as well as current situations within public cloud service providers in addition to private cloud storage applications.

The second portion of the thesis involves an implementation phase. The key component of the thesis will involve a complete instructions for building a personal cloud server via ownCloud. The implementing stage is comprised of the implementation plan and the working environment. It also involves the implementation phases, as well as testing and monitoring of the cloud solution.

## **2 RELEVANT TECHNOLOGIES**

This chapter considers the entire research and required tools and technologies review and summarise the implementation of the prototype and comparison of existing cloud service providers (CSP). It also reviews research papers and reveals the findings. A theoretical framework is also produced comprising the research concepts and theories. This will serve the readers to understand different CSPs and trends of cloud computing and the tools used in this research.

### **2.1 Cloud Computing**

Cloud computing has transformed how we store, access and manage data. According to Rice's (2023) detailed roadmap of cloud computing development, during the mainframe era, the concept of cloud computing was first discussed and developed with it the idea of linking computers together over great distances for a mixture of scientific and military purposes. This idea further shaped the development of cloud computing.

Cloud computing gives you on-demand access to servers, storage, databases, networking, software, analytics and intelligence, increasing innovation, providing flexible resources and saving you money. AWS, Microsoft Azure, as well as Google Cloud Platform, are major cloud computing providers. Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS) are among the many cloud services these companies offer.

Cloud services have been a popular service in this modern age. With the advancement in technology, everyone is using cloud services in one way or another, from uploading a single picture to storing large volumes of data, from connecting to another individual in another part of the world to expanding business over the internet. Cloud computing services are highly manageable, scalable, and adaptable, so they are popular among everyone, from single individuals to global corporations.

## **2.2 Consumer Choices of Cloud**

Studies of Americas Conference on Information Systems (AMCIS) looks at the increasing area of cloud computing, which is quickly growing and can change industry but does not have a solid definition and directs too much research to technical topics. Despite industry reports stressing the large benefits and meaningful challenges of cloud adoption, particularly regarding security and cost reduction, academic literature has largely neglected the consumer perspective, importantly concerning preferences and pricing strategies.

To explore aspects influencing cloud service adoption past simple technical elements, the research by Koehler et al. (2010) employs choice-based conjoint analysis to scrutinize all service attribute preferences of many small and medium enterprises (SMEs) throughout Singapore, intending to fill such void. Contrary to expectations, it (Koehler et al., 2010) also says that cloud provider selection isn't mainly driven by financial aspects like cost savings, the findings reveal. Instead, the cloud service provider's reputation emerges as a more important factor, along with a thorough adherence to standard data formats.

Specific consumer categories having different preferences are pinpointed via a thorough analysis incorporating both Hierarchical Bayes estimation as well as cluster analysis, which indicates several chances for second-degree price discrimination (Varian et al., 2004). AMCIS also describe regression analysis of tariff preferences that indicates important correlations with insurance and flexibility effects. This study (Koehler et al., 2010) is useful to practitioners, as it offers a consumer-centric perspective on cloud service attributes and their effect on service design and pricing. A great deal of research should assess how much people will pay to better understand how consumers act in cloud computing. A range of cost-benefit analyses of service features like versioning should also be performed.

## 2.3 Comparing CSPs

There exists a considerable body of literature on the study by Almishal and Youssef (2014) which compares the major CSPs in market, while discussing the characteristics of cloud computing along with the types of cloud services. Amazon, Google, in addition to Microsoft, are definitely among the most ubiquitous providers. These companies provide multiple cloud services, including infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS). The article concludes by discussing the possible future of cloud computing.

Table 1. Comparison of major CSPs.

| <b>Cloud Service Providers</b>       | <b>Amazon</b>                 | <b>Google</b>                                   | <b>Microsoft</b>                              |
|--------------------------------------|-------------------------------|-------------------------------------------------|-----------------------------------------------|
| Types of services provided           | IaaS, PaaS, Storage, Database | IaaS, PaaS, Storage, mobile, database, Big Data | IaaS, PaaS, mobile, Media, Database, Big Data |
| Cost for 100 GB storage monthly (\$) | 2.30                          | 1.90                                            | 2.08                                          |
| User friendliness                    | Good                          | Good                                            | Good                                          |
| Security level                       | High                          | High                                            | High                                          |
| Security certification               | Yes                           | Yes                                             | Yes                                           |

Cloud computing has become influential in the IT landscape as well as gaining attention in recent years because of benefits like reduced IT costs along with management. In addition, cloud storage and multiple social networks have transformed the IT social interactions and communications of countless people. The CSP capably and thoroughly manages the cloud, offering both expanded storage space and plentiful computational resource. The client or the owner is the individual or organization that buys or rents cloud-computing service for computation or for storing data. Each individual using the cloud service is a user.

## 2.4 Challenges and Benefits of CSPs

Cloud computing exerts a large effect on all information and communication technology and represents one of the foremost technical forces behind any enterprise digitalization. The uncertainty along with the risks for user companies have considerably increased in adopting cloud services. This is because of the escalating dissemination of cloud services as well as the surprisingly growing number of CSPs. According to Hentschel et al. (2018), there are several challenges faced by CSPs. Below are some of them:

- Customers frequently hold important concerns regarding data security, privacy, as well as data centres. CSPs have to promise full transparency and give secure solutions for all to get rid of every one of these worries.
- Customers may resist adopting cloud services if they fear losing control over their IT infrastructure and data.
- Vendor lock-in worries many customers, since it has the potential to occur if they depend on a single CSP to a great extent.
- Customers want solutions made for their needs, CSPs have to find balance between offering services that are standardized and offering solutions that are customized.
- The sales process has become importantly more complex. Explaining both technical and legal aspects of cloud services to customers is required.

These are some of the challenges that CSPs encounter in meeting customer expectations, building trust, and staying competitive in a rapidly changing market. They (Hentschel et al., 2018) offer a thorough overview of the issues that must be addressed concerning cloud service adoption and provider selection.

The primary advantages of cloud computing services are fundamentally connected with its essence. Bălăţescu (2014) enumerated five essential advantages of cloud computing services:

- On demand services – the client can access the services without interaction with cloud service provider personnel
- Broad network access – access to resources in the cloud is available over the network using standard methods in a manner that provides platform independent access to clients of all types
- Resource pooling – resources are pooled in a system that support multi-tenant usage.
- Rapid elasticity – resources can be rapidly and elastically provided.
- Measured service – clients are charged based on a known metrics such as amount of storage, number of transactions, bandwidth etc.

Additional benefits of cloud computing and its accompanying services can comprehensively provide access to a wide range of applications without downloads or installations, greatly lowered costs, complete ease of use, assured service quality as contractually agreed, fully outsourced IT management, simplified maintenance and upgrades, access to the application from any computer via the Internet, scalability via on-demand resources, and a pay-as-you-go pricing model.

## **2.5 Raspberry Pi**

The Raspberry Pi is often referred to as a Single Board Computer (SBC), meaning that it runs a full operating system and has sufficient peripherals to start execution without additional hardware. Although other SBCs existed before, the Raspberry Pi Foundation made the SBC accessible to almost anyone, introducing not just a low-cost computer, but one that can bridge the gap to the physical world via standard electronic interfaces; controlled programmatically from the operating system. This has led to the popularity of the Raspberry Pi, not only in education but with industry, hobbyists, prototype builders, gamers and



the curious, and it has enabled people to experiment in new ways (Simon & Steven, 2017).

The Raspberry Pi is indeed a powerful prototyping platform; even though the idea of a prototype suggests a somewhat partial implementation of every single desired feature, a need to build a completely functional prototype usually exists. A prototype should be built for a couple of genuinely important reasons:

1. To test rigorously and validate fully an idea or hypothesis. The fail fast design philosophy involves identifying ideas early. This means distinguishing between good and bad ideas. If a prototype offers a way to either prove or disprove the feasibility of an idea, building it in only a few days is acceptable, even if it is quite large, rather expensive, consumes a large amount of power, and is somewhat slow.
2. Hardware design can be validated. Before fully optimising a design or commissioning a large production run, a prototype can be used to validate the sensor capabilities and electronic design within a desired operating environment.

The Raspberry Pi is effective for this purpose since it runs favoured Unix-like operating system variants along with supporting high-level programming languages in addition to being commodity hardware.

## **2.6 ownCloud**

ownCloud is free and open-source software hosted in a data centre, on servers, and using storage that all belong to the user. ownCloud integrates smoothly with user IT infrastructure; therefore, it can furnish file sharing services that adhere to its data security as well as compliance policies, also rigorously maintaining data in its original location. It importantly provides several well-known cloud services. ownCloud's extensibility offers a large amount of strength. It offers different functions like

live document editing, photo galleries, calendars, many contacts, and plentiful methods of file syncing, so it can function as either a cloud-based data hub for local programs or a browser-based desktop. Since you can create and run many user accounts on a single instance, own-Cloud is a good solution for personal use, families, or even businesses (Rexha et al., 2012).

## **2.7 User Concerns**

Users today retain all their personal data in the cloud. Those data include various pictures, numbers and bank cards and many other private documents. Of course that would be very inappropriate and no one would like someone seeing things that he considers personal. In case of some clouds, the administrator or employees who have administrative access to the hosts (file servers) on which the drive data is stored will have read-access to user data: this includes the operations engineers, service engineers, system administrators: these type of employees are typically the ones with that type of access and the ones that can see user data and of course this is not preferable for anyone. The cloud can be a dangerous place, and Cloud apps are not without their flaws in this area. There have been incidents where user data was compromised or left unprotected on the service, and it has had extended outages. So the best would be to apply security not only during their transfer but also during their storage in the server.

## **2.8 Hardware requirements**

1. Raspberry Pi: A device which is used for low computing operations. It serves the purpose of a mini computer. It has Double band LAN, USB ports, HDMI port. It can be used as a server also.

2. Raspberry Pi power supply 5v/10v: To power up this device at least 5v-10v power supply is needed. A power supply adapter with appropriate pin type should be chosen.
3. Micro-SD card 4/8 GB class 10: MicroSD card serves the purpose of rom of the raspberry pi. Entire raspbian os is installed on this microSD card. For proper functioning of this device at least class 10 or U1 microSD cards should be used.
4. Internet connection (LAN/Wifi): Internet connection is a must in setting up the entire set-up as well as further using it. As all the commands to be executed need internet and further tasks to be performed too. A LAN or Wifi is recommended.

External hard drive: An external hard drive is needed to make it a centralised cloud storage. It needs to be mounted before using it. Then it needs to be linked with Owncloud so that the files stored in it can be accessed.

## **2.9 Software requirements**

1. Raspbian OS: It is an operating system which has been designed specifically the for Raspberry Pi device. It is based on Unix/Linux. It consists of Python, Scratch, Sonic Pi, Java and that's just beginning.
2. Linux terminal/Putty: Linux can be used to configure Owncloud server with the raspberry pi. For Windows users Putty can be used which is an open source terminal emulator. Otherwise Raspbian OS is based on linux so it has inbuilt terminal.
3. Constant internet connection: Internet connection is must for configurations and further using it. Owncloud would also need the internet connection for its work.
4. Application (Owncloud): Owncloud is suite for file hosting services. It is same as Dropbox but the distinction is that

owncloud does not offer data centre capacity. It is open source. Users don't have to pay any amount for their private servers.

With Raspberry Pi, an external hard disk is used for private cloud storage. The user is able to choose the amount of memory on their own. Also, they can have full power over their information. With ownCloud, they are able to get to the info on the external hard drive through any internet-connected device, and handle the external hard drive as if it was a cloud storage device. As information increases throughout time, storing it turns quite important because the number of important things relies upon the information. Procuring pricey storage tools and transporting them along turns quite difficult on the customer. That is where distributed storage gets more attention. Using a form of distributed storage means the client does not have to buy expensive storage devices or deal with the hassle during the transporting of them. Cloud specialist organizations give the distributed storage to many clients, as well as those clients can get to their information by using any gadget which has web empowered in it.

### **3 APPLICATION DESCRIPTION**

This project focuses on the creation of a personal cloud storage setup via ownCloud and Pi. The central idea presented here is to fully empower all users with a scalable and secure, efficient, plus budget-friendly cloud storage alternative right now. It tackles rising demands with regard to accessible as well as secure data management, most notably for people, small businesses, along with prototype creation. It digs into whether a private cloud is better than using CSPs.

One key aspect of the project involves showcasing the initial setup of a private cloud storage. Another key aspect involves managing a single private cloud server through ownCloud via a single Raspberry Pi. It offers exact instructions for building a cloud server, such as setting up, testing, and monitoring. The thesis virtually aims to show that an ownCloud-based private cloud server offers an integrated and also a secure cloud solution, particularly suitable for individual use. It also truly seeks to fully raise even further awareness regarding optimal cloud storage practices and fully address multiple data security and privacy concerns within cloud computing.

The thesis was designed as a mix of comparative and qualitative research in order to proof the thesis 's concept. The audience scope of the research is mainly home users and small scale business owners. The implementation part is building a private cloud server. The private cloud server will be built with ownCloud. The system is expected to give positive results in storing data in a private cloud server. Performing an experiment step by step strictly on a private cloud server is completely required in order to provide readers with a proof of concept that data security specifically in a private cloud is on a prominent level. The experiment will be the actual proof of concept. It will also be for readers who thoroughly desire to build along with to control of their own private cloud server.

### 3.1 QFD (Quality Function Deployment)

A QFD diagram of the proposed system is given below. It outlines the requirements for building the personal cloud based on priority.

Table 2. Requirements Analysis

| <b>Customer Requirements</b> | <b>Technical Requirements</b>                                                       | <b>Priority (1-5)</b> |
|------------------------------|-------------------------------------------------------------------------------------|-----------------------|
| Secure Data Storage          | SSL Encryption, User Authentication, Access Control, Regular Security Updates       | 5                     |
| Accessible from Anywhere     | Remote Access Configuration, Cross-Platform Compatibility, User-Friendly Interface  | 4                     |
| Cost-Effectiveness           | Use of Raspberry Pi, Open-Source Software (ownCloud), Efficient Resource Management | 5                     |
| Scalable Storage             | Expandable Storage via External Hard Drives, Efficient File Management              | 4                     |
| Easy to Use                  | Clear Installation Guide, Intuitive User Interface, Simple File Sharing             | 3                     |
| Reliable Performance         | Stable System Operation, Data Backup and Recovery, Minimal Downtime                 | 4                     |
| Data Privacy                 | User Data Control, Transparency, Compliance with Privacy Principles                 | 5                     |

To ensure clarity as well as academic rigor, the thesis involved in creating a personal cloud via a Raspberry Pi plus ownCloud may be prioritized as: Primary focus is ensuring safe data storage. Specific users must have such a total degree of confidence in the overall safety of their data, thus making the complete implementation of strong security protocols critical.

Of close importance is that of cost-effectiveness. One primary motivation for selecting a personal cloud solution is for reducing the overall

expenses associated with commercial cloud services, thereby making such a solution a vital requirement. Data privacy is always a top priority, propelled by users seeking command over their information. This drive comes from guaranteeing privacy protection, especially in contrast to the many concerns often linked to public cloud solutions.

Furthermore, the ability for access of data from multiple devices from and locations, or accessible from anywhere, is a key benefit expected from any cloud solution. Progress toward scalable storage is also quite important, as the capacity for expansion of storage becomes vitally important for long-term usability. Users do also prioritize very reliable performance, fully expecting quite consistent availability as well as highly efficient operation of the cloud storage. Usability, while still key, is viewed as less vital than security and cost savings. This occurs simply because users might tolerate a less refined interface if the solution delivers strongly enough in other key areas.

### **3.2 Functional Definition**

This paper defines the role of a private cloud server using a Raspberry Pi and ownCloud. The chief aim of this venture is simply to let user's setup a cloud that is private and self-run. This setting enables users to properly save, quickly reach, as well as easily handle their digital info, giving a better choice to usual shared cloud tools. Of the entire system, the design prioritizes user autonomy, total data security, as well as large cost-effectiveness.

The functional scope includes many key aspects thereof. To begin with, the platform offers multiple data storage features as well. The platform also provides wide-ranging access features for users. It lets those users keep many kinds of digital things, such as files, papers, pictures, and media, on a drive hooked up outside to the Raspberry Pi. Users can then retrieve and thus access this data remotely from virtually any location

with internet connectivity, granting to them additional flexibility and more control over their information.

User management abilities are also within the system. It eases greatly for the smooth creation and administration of user accounts, thereby enabling more secure and controlled access to all the stored data. This feature is key for keeping data quite private as well as for stopping access that is surely not allowed.

Thirdly, security really and truly is indeed a very necessary functional aspect. The system thoroughly and comprehensively implements a few security protocols targeted at fully protecting user data all throughout its complete lifecycle. These actions include SSL encryption ensuring secure data transit, user authentication verifying user IDs, and access control managing data access rights.

Also, the platform even accounts for the file management capabilities. Tools are available so users can take routine actions on files. These actions include uploading, downloading, deleting, renaming, and organizing files. This makes management of all digital assets efficient as well as user-friendly within personal cloud storage.

Fifthly, the very project includes, specifically in it, a functional aspect that is system configuration and management. The thesis absolutely provides a quite detailed procedure for properly setting up and carefully configuring the Raspberry Pi and ownCloud software to establish the private cloud server. It gives users the totality of the data and the directions which are needed for running their cloud solution completely by themselves.

Lastly, the complete system is designed quite well with full scalability in mind for improvement. It fairly allows for such a large expansion in the storage capacity via the swift addition of more external storage devices. This absolutely ensures that the system can change greatly. That system can change to address storage needs of users over time. In order to



summarize, the working definition presents such a scheme that values certain user power, total data security, and particular price efficiency. Using available hardware with open-source software, the project gives a functional personal cloud option to let users handle data efficiently and securely.

### 3.3 Use Case Diagram

The diagram below provides a high-level overview of the functionalities of the proposed personal cloud system and how a user interacts with it.

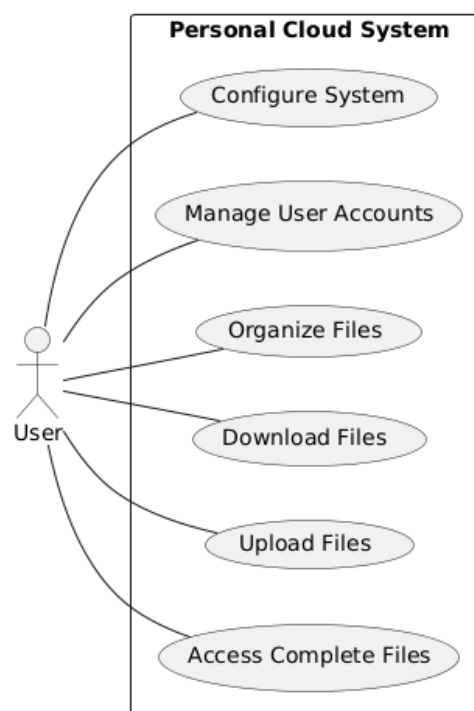


Figure 2. Use-case diagram

The use-case diagram shows graphically how the user interacts with the personal cloud system. As the main actor, the user interacts with the system variously. Functionally, the user can access complete files, which involves retrieving and viewing total stored data from each device connected to the system. The user also has the complete capability for uploading files, transferring data from those local devices to such personal

cloud storage. Conversely, the download files use-case allows users to retrieve data. It allows them right from the cloud storage down to their devices.

Furthermore, users can readily organize files within the cloud environment, performing actions such as creating many directories, renaming each file, and then managing the entire file structure. Management of user accounts is just another key use-case, enabling administrative functions like creating, modifying, and deleting user accounts, as well as managing user permissions.

In addition to direct data interaction, the user also engages in specific configuration of the system. This includes the setup as well as the customization of the personal cloud server, even including network configurations along with storage settings. Importantly, the configure system use-case also includes ensure security. This highlights that complete security configuration is an integral part of the system setup and its management. Hence, this diagram depicts the different manners in which a user engages with the personal cloud platform, spanning from file actions to the configuration of the system.

### **3.4 Class Diagram**

The architecture of the personal cloud storage setup, using a Raspberry Pi and ownCloud, features many key interrelated sections. At the very forefront indeed still stands the User. The user is that individual who engages with the system to store as well as retrieve data. Each client device, just like a mobile phone or computer, is indeed how this interaction happens, as well as being the interface to use cloud services. One network connection simplifies the communication between every client and server, which enables control signals and data transfer. The Raspberry Pi hardware functions as the system's central processing and storage hub. It hosts important cloud function software packages.

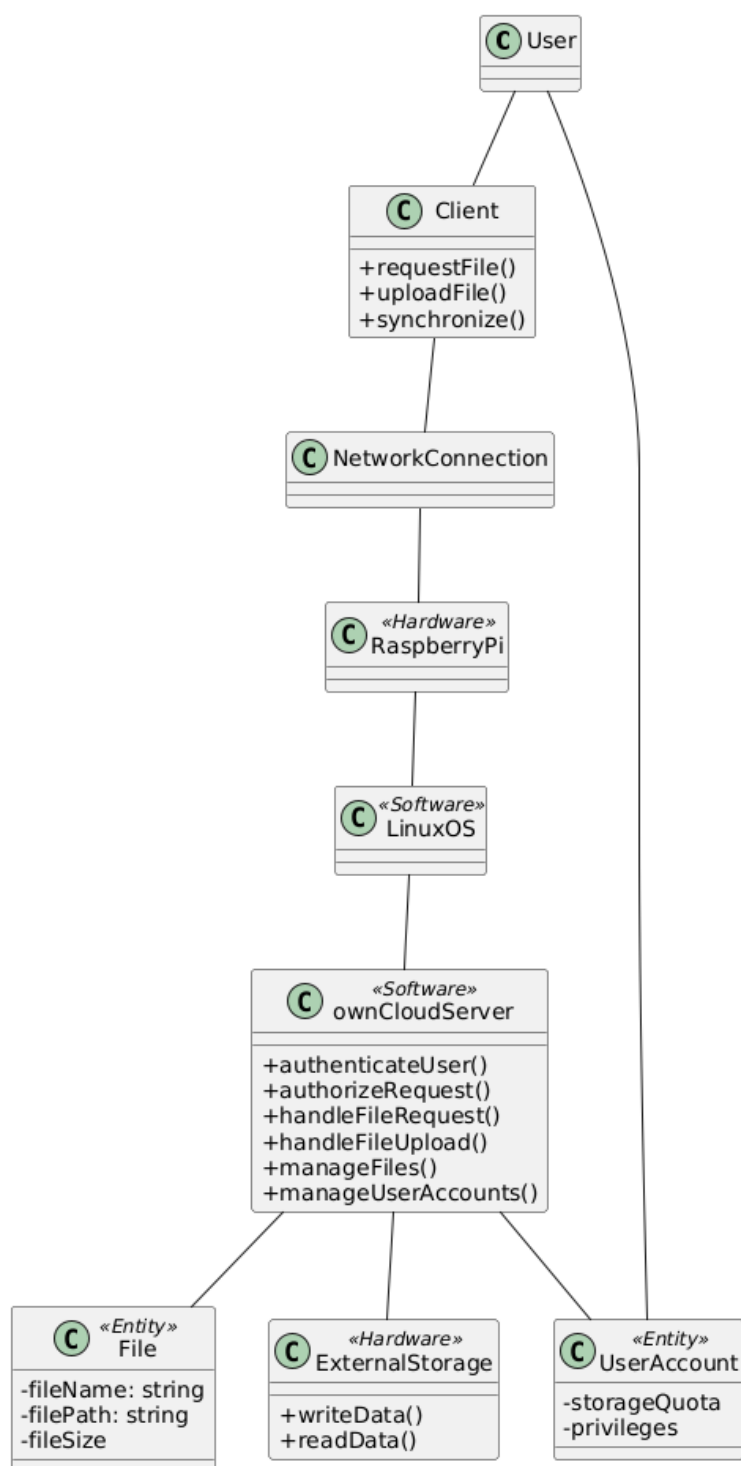


Figure 3. Class Diagram

Above figure is a class diagram for the personal cloud storage system using Raspberry Pi and ownCloud. This diagram focuses on the key software and hardware components and their relationships.

Functioning right on top of the Raspberry Pi is the Linux OS, thereby furnishing the fundamental environment truly for the running of applications and for managing of system resources. The ownCloud server, which is the key software, mainly sits upon this operating system as well as oversees the cloud storage features' implementation. The ownCloud server oversees every data storage for users inside file entities, which completely represent digital content kept throughout the system. To provide reliably safe and personalized access, the ownCloud server manages user account entities, defining user-specific quotas and privileges. By the Raspberry Pi, the physical storage for these files is handled via an external storage device, providing the necessary capacity for the data for the user.

Each user's experience starts as that client device asks ownCloud server via network connection. The client allows for the user to requestFile(), uploadFile(), and synchronize() their data with the server. ownCloud server deals with requests at just the server level, thereby performing such vital functions as authenticateUser() to even verify user ID as well as authorizeRequest() to ensure resources get proper access. The server then interacts completely and directly with the external storage so as to writeData() for uploads and to readData() for downloads, effectively managing the user's files.

The complete integrity and careful organization of all the stored data are maintained through the File entity, which tracks important metadata such as fileName, filePath, and fileSize. Each user account regulates system access, which is naturally tied in with each user as well as specifies parameters such as storage quota. The ownCloud server decidedly and effectively plays a key role in the thorough managing of both file entities and user account entities, thereby ensuring better data consistency and user access control. Fundamentally, the fluid connection of multiple hardware as well as software parts lets users make private cloud storage. Through this very connection, users do also use private cloud storage run by a Raspberry Pi and ownCloud.

### 3.5 Sequence Diagram

This sequence diagram displays a typical process for retrieving one file from the personal cloud storage system. We can create several similar diagrams for multiple use cases like uploading of a file or synchronizing of data, to greatly further understand all the interactions within the system.

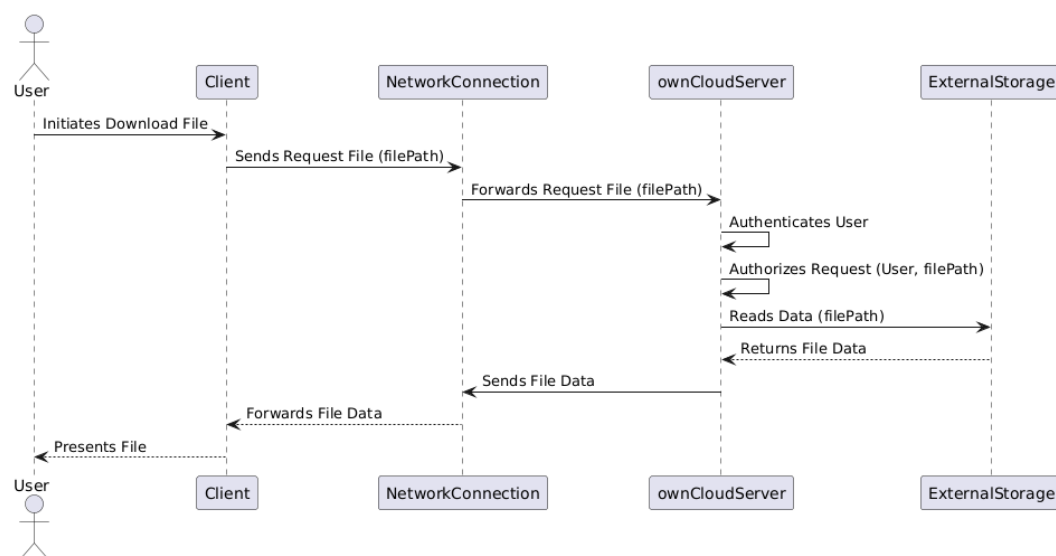


Figure 4. Sequence Diagram

To retrieve a file from personal cloud storage, via the "Download File" use case, the User, as primary actor, starts by requesting a file via the Client application. This action triggers several interactions between the system's core components to fulfil the user's request. Then, the Client will send a "Request File" message through the Network Connection to the ownCloud Server, including the exact filePath of what it wanted. This initial transmission signifies the start of some server-side processing, required in order to locate as well as deliver the requested data.

Upon the receiving of the file request, the ownCloud Server, the central orchestrator of the system's entire functionality, undertakes important security and access control measures. First, a complete authentication happens, verifying the exact identity of the user requesting, so fully legal users access all data stored. When authentication succeeds, the server

authorizes, verifying the authenticated user has access to the specific file. These checks are important and primary. They help in maintaining the full security and privacy of the user's stored information in the personal cloud environment.

The user's successful authentication and authorization is presumed. The ownCloud Server then works with External Storage, the user's files' physical repository. It sends one "Reads Data" request to the storage device. It carefully specifies the filePath of the file to be retrieved. The External Storage responds by completely locating each requested datum, in addition to fully transmitting the "File Data" back toward the ownCloud Server. This data retrieval highlights how external storage is majorly involved in providing content user's request.

Once the ownCloud Server gets the "File Data" from storage, it gets ready to send it to the client. The "File Data" goes directly across Network Connection, thus reversing initial request's usual path. Finally, the Client absolutely receives each of the transmitted "File Data" and then undertakes the core task of presenting all that data to the User. This presentation may exist as multiple types, like starting a file download to the local user device, showing the file's content inside the client app, or enabling the file for more interaction. This entire series of different actions fully shows exactly how the user, specific client, network, remote server, and extra storage parts all work jointly together to easily allow the central key feature of properly retrieving diverse files inside that personal cloud storage system.

### **3.6 Component Diagram**

The component diagram of the personal cloud storage system gives a structural overview over its primary software and hardware elements with their interrelationships. The system is broadly divided into both the Raspberry Pi Node, which hosts the server-side logic as well as data storage, and also the Client Device Package, representing the user's own

interface. This architectural representation stresses the modularity as well as dependencies natural in the system's design, easing a clear comprehension of its running framework.

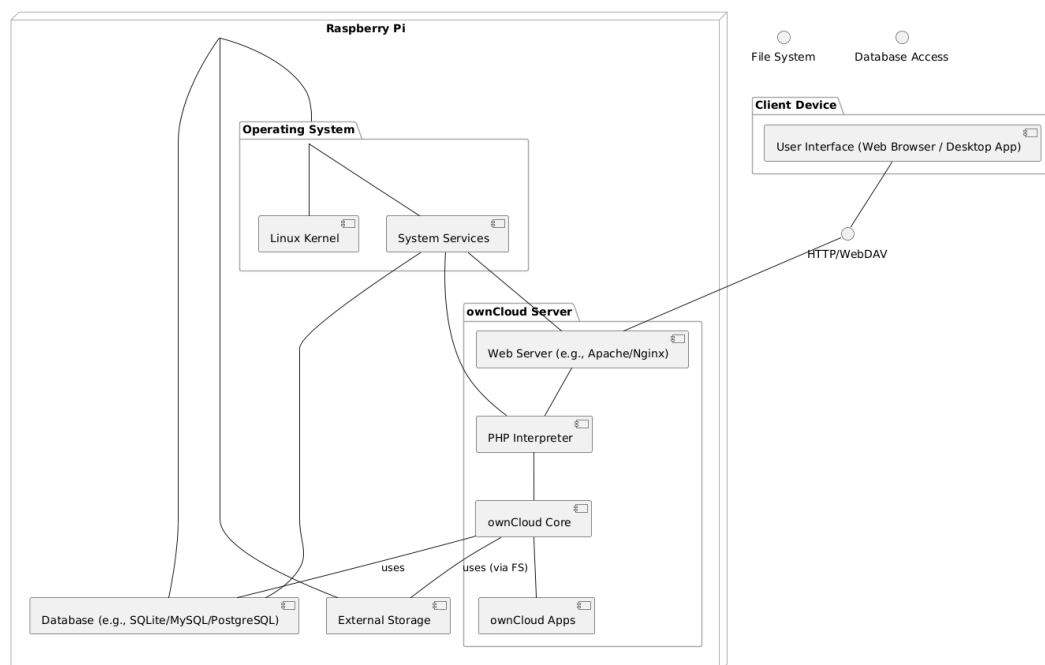


Figure 5. Component Diagram

Specifically, within the Raspberry Pi Node, the entire foundation is laid by the central Operating System Package, fully comprising the Linux Kernel, specially responsible for all core system functionalities, and System Services, which also provide important background processes. Building upon this foundation is the ownCloud Server Package, at the heart of the cloud storage application. This package includes the Web Server, which manages client requests via HTTP, along with the PHP Interpreter, which executes the ownCloud Core and its associated ownCloud Apps, extending the system's capabilities. Persistent data storage is managed by a Database. This Database stores metadata, and the External Storage stores the physical medium for user files.

The Client Device Package features the User Interface (Web Browser / Desktop App). It is serving as the primary point of interaction for the user. This interface fully communicates with the server-side components

through the usual HTTP interface, enabling the efficient transmission of requests and data. On the server side, the ownCloud Core certainly uses the Database Access interface to interact quite closely with the database and also the File System interface to manage each of the files on the External Storage. These interfaces abstract the underlying details of implementation, allowing for modularity. Flexibility exists within the system.

The linked dependencies among these components thoroughly highlight the overall flow of data and control. The ClientUI initiates communication through HTTP toward the WebServer, which in turn delegates processing unto the PHP Interpreter. The PHP Interpreter executes all of the ownCloud Core, which in turn leverages some ownCloud Apps for a few extended features. The ownCloud Core quite largely relies upon the Database specifically for metadata management purposes, and the External Storage particularly for file storage operations, all eased through the respective software interfaces. The full server-side function is backed by the OS, with System Services giving key support to the Web Server, PHP Interpreter, and Database. Ultimately, all these software components are deployed as well as executed on the Raspberry Pi hardware, which is physically connected to the Database (if local) and also the External Storage, thereby forming the complete personal cloud storage ecosystem.

### **3.7 Architectural Diagram**

The architectural diagram depicts very clearly a model for that kind of personal cloud storage apparatus, all focused on one Raspberry Pi board that is working like the server. This self-hosted solution integrates the Raspberry Pi with an external hard disk drive, providing sufficient storage capacity for personal data. The principal software component being depicted is ownCloud, running also on the Raspberry Pi, and it manages all of the cloud storage functionalities. The diagram additionally indicates





## 4 IMPLEMENTATION

A blog by Hashemi-Pour et al. (2025) says that cloud server is called private cloud when the cloud infrastructure operated for a single organization, managed internally or by a third-party and hosted internally or externally. The private cloud is created, maintained and managed by a single organization. Moreover, the private cloud infrastructure and resources might be available in the organization's data center (on-premises) or in a separate infrastructure (off-premises) (Hashemi-Pour et al., 2025).

The cloud solution that will be produced is a private cloud server inside the facility. Compared with several accessible SaaS, PaaS, and IaaS solutions within a marketplace, the experiment not only shows a manner for building a private cloud server solution but also gives a comprehension into many layers needed inside each cloud service. People are accountable for every layer around it, like the physical environment, network, infrastructure, operating system, user access, data and application.

Since the thesis is a proof of concept research, a detailed description of building a private cloud server is given to prove the research's hypothesis. The experiment is divided in three phases which are: installation, testing and monitoring. The research starts by getting the equipment together. This involves the acquisition of precisely one Raspberry Pi, one strong power supply unit and one high-capacity storage device, such as a single external hard drive or a single solid-state drive (SSD) and their smooth integration. Superior connectivity is guaranteed by incorporating one or more important networking peripherals: an Ethernet cable or a Wi-Fi adapter, depending on your preferred network connection.

#### 4.1 Implementation of cloud server

A cloud server is powerful physical and virtual infrastructure that performs application and information processing storage. Cloud servers are made using virtualization software, which splits a bare metal server into multiple virtual servers. Cloud storage is a cloud computing model where the data is stored across multiple remote servers as well as maintained via a cloud storage service provider. This allows users to customize their details and share them with contacts and colleagues on the Internet. In conjunction with the fact that I have showcased this project in the local network, port forwarding is not an option to consider now.



Figure 7. Raspberry Pi 3b.

From the above diagram we can see that the board has only precise components and also compact and portable. Raspberry pi is used as a central processor, in this project raspberry pi 3b model is used. We can also change the specification of the processor such that if we are using this processor for the hospital or huge organisations we can use higher level processor and if we use processor for schools or small level organisations we can use lower level processor as just the efficiency of the processor varies.

The normal power usage of the power adapter is around 5V/2A but if we use more storage space by adding three or more harddisks this power

is not sufficient to run the cloud so the power range varies a little amount of 5V/3A in order to run the cloud in a more efficient manner.

Here in our project we are using one internal harddisk of around 500 GB which will be sufficient to run using 5V/2A and if we need to increase the storage it is possible to add additional internal harddisk just the power range varies to 5V/3A but it can also be possible in such a way that if we use only one internal harddisk of higher storage it will be more compact and more efficient to run using 5V/2A and it is a life time investment and we can easily add or remove the additional storage at the user convenient.

We have to connect the ethernet cable to the raspberry pi because the cloud storage will only be active only if we supply endless amount of fast internet though the raspberry pi has inbuilt wifi it is not that much sufficient to provide fast internet so we have connected the raspberry pi to the ethernet cable for much more efficiency.

## **4.2 Installation and set up**

To create a personal distributed storage, a Raspberry Pi along with an external hard drive were required so that it could be converted into a cloud storage, alongside a mini-SD card that worked as ROM for the Raspberry Pi. OwnCloud is the platform used for getting to the cloud storage, for managing data inside, and for putting data into or taking data out of the cloud storage. Notwithstanding a few other existing file hosting services, such as Dropbox, Google Drive, as well as iCloud, an individual could create their very own cloud storage space along with use ownCloud as a secured avenue for access to their cloud storage drive.

The Raspberry Pi acted as a server, but it was previously configured for saving multiple data items kept from ownCloud. To achieve this specific

goal, ownCloud should be installed directly on the Raspberry Pi. The cloud storage involves an external hard drive.

To set up a Raspberry Pi, the initial needed step to readily take is to make Raspbian smoothly run inside it. After the full Raspbian disk image file is downloaded, it must be fully decompressed. After that, put Raspbian OS on a microSD card by hooking it up to a computer and using a tool like Win32 Disk Imager. After Raspbian is fully as well as completely installed onto the microSD card, carefully put it into the Raspberry Pi. When beginning with the device, the username "pi" together with the password "raspberrypi" may be used. This is just for the login purposes.

To set up an ownCloud server right on the Raspberry Pi, the start of the process involves using Pi's terminal, or a Linux terminal, with Putty efficiently working as a helpful resource for Windows users. Subsequently, the setup utility is often chosen to put multiple changes into action. These changes include the setting of the locale to en\_US.UTF8 in internationalization options, the adjustment of the memory split to 16m in advanced options, as well as overclocking to medium. The following step involves fully adding the www-data user to the entire www-data group. After that, all required ownCloud execution files, such as php7.2 and its modules, are set into use.

Subsequently, an SSL authentication is made through several stages between the website possessor and server, with defined procedures giving the website an SSL Certificate. This authentication, important to the maintaining of user together with site security along with reputation, necessitates, during the whole process, generating a custom DH Parameter record for the securing of SSL connections; in addition, the overall process may take as long as 16 hours, though adding the -dsaparam flag can expedite the whole process at the cost of the reduced security. After the generation, the three cert records go through chmod. The

server setup record is extracted so a user can enter their version. Ultimately, the server setup is arranged. This is for guaranteeing ownCloud operates as it should.

The subsequent step attentively requires coding to efficiently set up the Nginx web server, and that strictly means swapping out "technet.example.com" with the Raspberry Pi's local IP address or the domain name, absolutely making sure it accurately matches the info attentively given when carefully signing in, or ownCloud simply won't work. Following that, the user should entirely save and then quit. For adjustment to further settings, it requires opening of the PHP config. The user must then locate as well as update specific lines with care, using Ctrl + W in order to search, before saving along with exiting thereafter.

Next up, the dphys swapfile needs to be edited by the user, specifically by modifying a certain `conf_swapsize` line inside. Once these steps are complete, the Raspberry Pi must be restarted. ownCloud should be installed onto the Raspberry Pi. This should be done after the Pi restarts. To apply prior modifications, the `.user.ini` file must be opened after installation. Within this document, the user must revise accompanying qualities to 2000M. Upon completion, ownCloud must have the ability to link to the Raspberry Pi's IP address.

To mount and configure on a hard drive, you should begin by installing into an NTFS bundle if it is an NTFS drive. Following this action, make a directory in order to mount the drive on. After that, locate the GID, UID, and UUID. Finding the UID is a rather key thing to do. Likewise, finding the precise UUID for the hard drive allows the device to recognize that hard drive, even if it is connected at a different spot. You should, without a doubt, put your drive in the current `fstab` entry to completely guarantee it starts up using the proper permissions.

By adhering to these exact, specific steps, update the indicated UID, GID, and UUID with the precise values from which you completely got. Finally, reboot the Raspberry Pi device, as well as after that, the hard

disk should mount. The setup is finished. It is complete when the disks are mounted on.

To set up Owncloud on a Raspberry Pi, the initial action is to input the Raspberry Pi's IP address into the program. In the event a login issue arises when reaching the IP, simply account for it as an exception to proceed. When Owncloud is opened, a basic setup screen that is free of errors should show up. Enter for the username and also the password. Then, click on storage in addition to database, plus input the external drive path into `"/media/owncloudrive"`. After fully completing these steps, carefully select final finish.

## 5 TESTING

Testing constitutes a key part of creation of a personal cloud storage system, which is the system using ownCloud and Raspberry Pi. The aim for testing is not solely to find problems, but rather to ensure the system functions fully in accordance with the intention and also meets the design goals. In this chapter, we will explore all of the importance in the testing of assuring data integrity, security, reliability, and the ideal user experience. Each of those factors plays a vital role in the determining of how effectively the cloud storage system will perform under real-world scenarios. Strict system testing helps us find vulnerabilities that could weaken the functionality of our cloud storage or expose user data. The testing phase aims for confirmation that the system operates smoothly, handles data accurately, as well as provides a secure environment for users.

The importance of testing cannot be overstated for cases, particularly for a cloud storage platform dealing with sensitive personal data. Data integrity is important for fully ensuring that users can trust that their files are stored accurately as well as retrieved without errors. Certain discrepancies may lead to data loss; the loss can severely impact certain users. Another critical area is security; user data must be protected by a personal cloud storage system from breaches and access that is forbidden. Testing at regular intervals can help to identify vulnerabilities within the system, allowing developers to address such issues before these may become meaningful problems. Likewise, reliability matters, since users need consistent cloud service operation without much downtime or many failures. Finally, a great user experience is important in user retention; quite simple and effective testing can help refine such interface with features, therefore making that service quite user-friendly with efficiency.

Backup and recovery testing forms an important part of testing cloud storage. This involves validating if the backup process runs as intended



and also ensuring that user data can be restored accurately when loss occurs. In the specific context of ownCloud and Raspberry Pi, backup testing generally may include creating backups of multiple various file types and ensuring they are saved correctly in both local and remote locations. Recovery testing can involve simulating a data loss scenario and evaluating just how effectively the system can restore files toward their original state. Strict testing gives users peace of mind and lets developers find areas needing improvement. It is necessary for developers to regularly update their backup as well as recovery processes, as changes to the system itself and to the storage methods can affect their effectiveness.

In summary, testing is key to ensuring that a personal cloud storage system's functionality, safety, and reliability is achieved on ownCloud and Raspberry Pi. Through quite thorough testing, developers can fully ascertain that their own system delivers to its stated promises of data integrity, security, plus a smooth user experience. Backup and recovery testing is of importance, as it prepares the system for handling data loss scenarios with grace. The total outcomes of the testing phase will, in the end, improve the quality. This improvement will concern the cloud storage service. Thus, thorough as well as continuing testing should be a part of any development cycle in order to ensure continuous improvement and user satisfaction.

## **5.1 Test Cases**

Testing software is important for ensuring it runs smooth and meets with user expectations. One task found inside each digital platform involves uploading files. In this scenario, we focus on the process for testing of uploading a small text file to the ownCloud web interface. The procedure involves logging in, clicking the upload button, in addition to selecting a file smaller than 1MB prior to confirming the upload. A test case that succeeds shows the file in the user's account after the process

for the upload. This action confirms the upload feature works and is an important base for users storing documents with the system. Performance in this area is generally quite straightforward, but ensuring users encounter absolutely no errors during the upload improves their entire overall experience of the platform.

User authentication is truly a vital aspect to all apps that are in need of protected access. The testing of login process via several attempts for logins having incorrect credentials serves to highlight robustness of security protocols that are in place. In this particular test case, the user enters a username that is indeed correct, but a password that is actually incorrect. When a user attempts to log in, the attempt must fail, in addition to an error message stating the credentials are incorrect. This mechanism has a certain vital role in the protection of user accounts. It is protecting each of those user accounts against access that is forbidden. By precisely recording the totality of results derived from the test case, developers are able to fine-tune the login process and make sure that users feel secure using the service. Error handling offers a meaningful role in this situation, as specific communication about login failures can guide users toward fixing their mistakes without frustration that is common.

Another important aspect for software testing is measuring performance, particularly in the handling of larger files. In the performance test case, the goal is to download a file, close to 1GB in size. Users log in to the web interface as they find a large file to initiate the download process. The anticipated result is that the download will start quickly and then completes free of errors. Observing the time that it takes for the download to both begin and then finish is key for assessing the efficiency of the service. User dissatisfaction can arise from delays or interruptions. Therefore, it is important to ensure the system smoothly handles large uploads and downloads. After the test, meaningful performance issues can be highlighted and addressed, for improvements in processing speed and user experience.

The conducting of thorough testing on file upload functionalities, user authentication, as well as performance metrics is paramount in ensuring of a reliable and efficient user experience. Users gain reassurance about functionality by confirming that only a small text file is able to be uploaded. Therefore, they receive a degree of reassurance of functionality. Also, stressing the importance of secure login processes protects each user's information, and performance testing helps in maintaining effective service delivery of large files. Every comprehension derived from the test cases works to validate the software's state and direct all upcoming improvements. With continual adjustments, plus incremental improvements, developers can provide a secure environment, user-friendly, conducive to productivity with trust among users.

## **5.2 Results of Test Cases**

In the area of software development, testing is vital, ensuring that the program works as it should. Implementing test cases allows developers to examine aspects within any system, including functionality to security and performance. The ownCloud platform, a cloud storage service used by several, was put through tests with various scenarios. These tests evaluated the ability for the upload of a small text file as well as the system's response to incorrect login details plus performance in the downloading of a large file. We gain certain valuable understandings into the system's reliability as well as its security by analyzing each test case's results. These understandings are vital for users dependent upon that platform for file storage needs.

When testing of the upload of a small text file, the process was straightforward as it was intuitive. The user logged into the ownCloud web interface, and then clicked the single "Upload" button. The user selected a text file, and the file's size was below 1MB. We expected the selected file would upload and appear within the user's files. This particular test case confirmed the upload function worked upon its

successful completion. The interface gave clear feedback, enabling users to understand that their file was being uploaded, as well as to confirm its successful saving. This positive outcome strengthens to a degree the tested reliability of the user interface, which is necessary for continued user engagement and satisfaction.

The second test case looked at system security, mainly at how it answered incorrect login attempts. A failed login attempt was expected, with an appropriate error message, after entering a valid username, but an incorrect password. The outcome remained as expected; the system denied all access as well as displayed a clear notification to indicate that the credentials were in fact incorrect. This result remains important as it clearly shows the effectiveness of the authentication mechanisms that are in place, further ensuring that forbidden users cannot gain simple access to sensitive information. This aspect of testing highlights the importance of security protocols. These measures protect user data, aligning with best practices in cybersecurity.

Lastly, the performance test involved the action of downloading of a large file, specifically one weighing around 1GB. This scenario targeted to thoroughly assess how well the ownCloud platform performs under stress, particularly during the transfer of greatly large files. After successfully logging in, as well as locating the file, clicking the download icon initiated the process. The foreseen result involved download commencement at once, and file transfer completion, without any errors, inside a suitable timeframe. The test case revealed that the download functioned effectively, allowing the file to be downloaded without undue delays and completing the transfer as expected. This outcome highlights the platform's power to manage large files, providing user satisfaction via a fluid experience. In general, the outcomes from all of these tests give a view of the ownCloud system's functionality and also security, plus performance, showing its reliability for each of the users.

### 5.3 Possible Improvements

To analyse test results effectively, one must focus on key aspects showing how users engage the system and pinpoint areas needing enhancement. The early test results indicate several functionalities of that system, several security protocols now in place, and its overall performance. We can gather understandings into how well the system meets user expectations by examining these results. Adjustments might be necessary in specific areas. A quite detailed review of these tests, in totality, will allow us to better understand user experiences, as this guides improvements that make the system efficient and user-friendly.

One critical element from the test results is how important user experience is. The assured upload of each file onto the cloud shows such system's ability for handling functional aspects reliably enough. This outcome reassures users seeking a handy way to keep data. In security, the test highlighted how the system prevents access that is forbidden. It does this by blocking any login attempts when a password that's incorrect gets used, reinforcing all safety considerations. Both of these areas contribute quite positively to user experience, showing how the system prioritizes functionality as well as security. Nonetheless, examining user feedback, along with these test results, could provide a detailed perception of certain experiences when using certain features.

While the test results point to skills in main functions, they hint at spots where the system needs work. Although uploads and downloads occur with no delays, user expectations demand greater performance levels, particularly for larger files. To improve download speeds, or to integrate more intuitive features for file uploads, could elevate user satisfaction considerably. While current security protocols work, stronger mechanisms like multi-factor authentication may be integrated. Thus, leveraging many understandings that come from these results will allow for data-driven recommendations about system improvements, in addition to aligning with user needs plus industry standards.

In conclusion, analysing of the test results is a step toward understanding how a system is performing from the user's perspective. We can recognize trends that assist in pinpointing areas requiring enhancement by focusing on functional reliability. Also, we can focus on security protocols and overall performance. Taking actionable steps which are based on these understandings will not just improve user satisfaction, in addition to strengthening the integrity as well as reliability of the system. This particular cycle involving testing, analysis, and improvement will ensure the system evolves to meet the changing expectations of users and continues to provide a user-pleasant interaction for them.

## REFERENCES

1. Almishal, A., & Youssef, A. E. (2014). Cloud service providers: A comparative study. *International journal of computer applications & information technology*, 5(2), 46-52.
2. Băltătescu, I. (2014). Cloud computing services: benefits, risks and intellectual property issues \*. *Global Economic Observer*, 2(1), 230–242.
3. Hashemi-Pour, C., Bigelow, S. J., & Lutkevich, B. (2025, January 31). *What is a private cloud? Definition and examples*. Search Cloud Computing. Retrieved March 11, 2025, from <https://www.techtarget.com/searchcloudcomputing/definition/private-cloud>
4. Hentschel, R., Leyh, C. & Petznick, A. Current cloud challenges in Germany: the perspective of cloud service providers. *J Cloud Comp* **7**, 5 (2018). <https://doi.org/10.1186/s13677-018-0107-6>
5. Jansen, W., & Grance, T. (2011). *Guidelines on security and privacy in public cloud computing*. <https://doi.org/10.6028/nist.sp.800-144>
6. Jena, S. R., Saini, D., & Sabharwal, M. (2020). Personal Edge Cloud using Raspberry Pi. *Journal of Xidian University*, 14(5). <https://doi.org/10.37896/jxu14.5/098>
7. KOEHLER, Philip; ANANDASIVAM, Arun; and MA, Dan. Cloud Services from a Consumer Perspective. (2010). *Proceedings of the 16th Americas Conference on Information Systems (AMCIS 2010)*, Lima, Peru, August 12-15, 2010. 1-11. [https://ink.library.smu.edu.sg/sis\\_research/1313/](https://ink.library.smu.edu.sg/sis_research/1313/)

8. Kumar, S. P., Rayavel, P., Anbarasi, N., Renukadevi, B., & Maalini, D. (2021). Raspberry pi based secured cloud data. *Journal of Physics Conference Series*, 1964(4), 042101.  
<https://doi.org/10.1088/1742-6596/1964/4/042101>
9. Ngo, T. (2018). Cloud Security: Private Cloud Solution with End-to-end Encryption. In *Degree Programme in Business IT* (pp. 1–60) [Thesis]. <https://www.theseus.fi/bitstream/handle/10024/154130/Cloud%20Security%20-%20Private%20Cloud%20Solution%20with%20End-to-end%20Encryption.pdf?sequence=1&isAllowed=y>
10. Nida, Pinki, Harsh Dhiman, Shahnawaz Hussain, 2014, A Survey on Identity and Access Management in Cloud Computing, *International Journal Of Engineering Research & Technology (IJERT) Volume 03, Issue 04 (April 2014)*.  
<https://doi.org/10.17577/IJERTV3IS040880>
11. Rexha, B., Likaj, B., & Lajqi, H. (2012). Assuring security in private clouds using ownCloud. *ijacit. com*.
12. Rice, I. (2023, June 3). Timeline of cloud computing.  
[https://timelines.issarice.com/wiki/Timeline\\_of\\_cloud\\_computing](https://timelines.issarice.com/wiki/Timeline_of_cloud_computing)
13. Sheps, A. (2024, July 23). *Top 10 cloud attacks and what you can do about them*. Aqua. Retrieved March 1, 2025, from <https://www.aquasec.com/cloud-native-academy/cloud-attacks/cloud-attacks/>
14. Simon, J. C., & Steven, J. J (Eds.). (2017). *Raspberry pi technology*: MDPI.



- 15.Suryawanshi, N. R., Amancha, N. R., Sambulkhani, N. S.,  
Shinde, N. V., & Vyavahare, N. P. (2023). Personal Cloud Storage using Raspberry PI. *International Journal of Advanced Research in Science Communication and Technology*, 13–19.  
<https://doi.org/10.48175/ijarsct-9465>
- 16.Varian, HR., J. Farrell, and C. Shapiro. The Economics of Information Technology. Cambridge: Press Syndicate of the University of Cambridge, 2004.